

Geometric Closure of the Design Space: The Missing Chemical Classes, the Scope of the SCE Framework, and the Programme for What Remains

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Part of the SCE Framework preprint series. Preprint 1 (foundation and fixed point): same repository.
This paper is the series capstone: it maps the framework’s reach, its limits, and the complete programme of
what remains.

Abstract

The SCE Framework (Preprints 1–5) establishes $SCE^* = 1.466$ as the fixed point of Li^+ first-shell coordination entropy that simultaneously maximises room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) in lithium metal batteries. Five preprints derive four candidate formulations and one arctic-optimised design, covering three orthogonal navigation mechanisms and one temperature-responsive design principle.

The present paper closes the design space geometrically. We systematically enumerate the chemical classes that are *absent* from the confirmed dataset and the series candidates, derive pre-registered SCE predictions for each class using the framework equations and molecular structure arguments, and establish the conditions under which each missing class is predicted to reach, approach, or be unable to reach SCE^* .

The missing classes identified and treated are: (1) nitrile-class solvents (acetonitrile, propionitrile), (2) sulfolane and sulfone-class solvents, (3) fluorinated ether sub-classes (HFE, BTFMD, HFTHP), (4) carbonate-class solvents in ether mixtures, (5) ionic liquid co-solvents, and (6) solid-state and quasi-solid electrolyte analogues.

For each class we derive: the expected SCE position relative to SCE^* , the mechanism by which it can or cannot reach SCE^* , the specific formulation prediction where applicable, and the falsification conditions.

We also derive the framework’s three fundamental scope limits: (A) the *coordination geometry requirement* (classes where Li^+ has no accessible coordination geometry distinction cannot navigate by the geometric mechanisms), (B) the *reductive stability requirement* (classes that are reduced at Li metal potentials before SCE can be established are outside the framework’s operating domain), and (C) the *concentration window requirement* (classes where SCE is concentration-invariant and $SCE \neq SCE^*$ at any accessible concentration are geometrically unreachable by concentration tuning alone).

The paper concludes with the complete novelty stack of the series, the full geometric structure of what has been derived and what remains, and a summary of the programme for the next generation of experimental and computational work.

The derivation record and all pre-registered conditions are archived at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

Keywords: lithium metal battery, electrolyte design space, coordination entropy, SCE*, missing classes, nitrile, sulfone, fluorinated ether, carbonate, ionic liquid, scope limits, geometric closure, programme

Contents

1	Introduction	4
1.1	What the Series Has Established	4
1.2	The Purpose of Geometric Closure	4
1.3	The Molecular Search Closure	5
1.4	Structure of This Paper	5
2	Missing Class 1: Nitrile-Class Solvents	5
2.1	Why Nitriles Are Absent from the Confirmed Dataset	5
2.2	SCE Position of Nitrile-Class Solvents	5
2.3	Can Nitrile/Ether Mixtures Reach SCE*?	6
2.4	Pre-Registered Prediction	6
3	Missing Class 2: Sulfolane and Sulfone-Class Solvents	6
3.1	Why Sulfones Are Absent	6
3.2	SCE Position of Sulfone-Class Solvents	7
3.3	Predicted Mechanism: Ordering from Above	7
4	Missing Class 3: Fluorinated Ether Sub-Classes	7
4.1	HFTHP and BTFMD: The MU Session Findings	7
4.2	HFTHP: Arctic Candidate B (Pre-Registered Prediction)	8
4.3	BTFMD: Intermediate Fluorinated Ether	8
5	Missing Class 4: Carbonate-Class Solvents in Ether Mixtures	9
5.1	Why Carbonates Are Absent	9
5.2	SCE Position of Carbonate-Class Solvents	9
5.3	The SEI Chemistry Complication	9
6	Missing Class 5: Ionic Liquid Co-Solvents	10
6.1	Why Ionic Liquids Are Absent	10
6.2	Framework Applicability to IL-Containing Systems	10
7	Missing Class 6: Solid-State and Quasi-Solid Electrolyte Analogues	11
7.1	Framework Applicability	11
7.2	Quasi-Solid and Gel Electrolytes	11
8	The Three Scope Limits of the SCE Framework	12
8.1	Scope Limit A: The Coordination Geometry Requirement	12
8.2	Scope Limit B: The Reductive Stability Requirement	12
8.3	Scope Limit C: The Concentration Window Requirement	12

9	The Regime Boundary and the SCE Band	13
9.1	The Band: Key New Result from the MU Session	13
9.2	Regime Boundary Position	13
10	The Complete Novelty Stack of the Series	13
10.1	Novel Theoretical Results	13
10.2	Novel Candidate Formulations (Pre-Registered)	14
10.3	Novel Experimental Predictions (All Pre-Registered)	15
11	The Programme for What Remains	15
11.1	Experimental Priorities	15
11.2	Computational Priorities	15
11.3	Theoretical Extensions	16
12	Discussion: The Series as a Complete Scientific Programme	16
12.1	What Has Been Achieved	16
12.2	The Nature of What Has Been Built	17
12.3	The Honest Assessment of What Remains Unknown	17
12.4	The Repository as the Geometric Backbone	18
13	Conclusion	18
A	Complete Class Status Table	19
B	The Web Structure of the Series	20

1 Introduction

1.1 What the Series Has Established

The SCE Framework series has, across five preprints:

1. Derived $\text{SCE}^* = 1.466$ as a fixed point of combined RT and LT performance from a 21-system dataset ($R^2 = 0.828$, $p = 4.5 \times 10^{-6}$).
2. Established three orthogonal mechanisms for navigating to SCE^* : ESP matching, anion reception, and within-class geometric diversity.
3. Derived one temperature-responsive design principle: the arctic optimum.
4. Specified six candidate formulations with pre-registered falsification conditions.
5. Confirmed the fixed point against six published papers; none falsifies the framework.
6. Mapped the molecular design space within the ether class (MU Molecular Search, same repository) and achieved geometric closure of the ether-class search.

What the series has *not* established is equally important: the framework has been derived from and applied to a dataset dominated by ether-class and phosphate ester solvents. The confirmed dataset contains no nitrile-class, sulfone-class, or ionic liquid systems. The fluorinated ether subclasses are represented only indirectly. The carbonate class is absent. Solid-state analogues are not addressed.

This paper closes the design space by addressing these absences systematically.

1.2 The Purpose of Geometric Closure

A scientific framework is not complete until its scope is defined. Scope definition requires two things: knowing what the framework covers (the domain of confirmed application) and knowing what it does not cover and why (the scope limits).

The SCE Framework’s scope is defined by:

- The coordination geometry requirement: the mechanism operates on Li^+ first-shell geometry. If a solvent class does not produce geometrically distinct coordination modes under mixing, the mechanism does not apply.
- The reductive stability requirement: the mechanism operates at Li metal potentials (≈ 0 V vs. Li/Li^+). If a solvent class is reduced before the electrolyte can form a stable solvation structure, it is outside the operating domain.
- The concentration window requirement: the framework’s dataset spans LiFSI concentrations of 0.5–2.0 M. For classes where SCE is concentration-invariant and $\text{SCE} \neq \text{SCE}^*$ at all accessible concentrations, concentration tuning cannot reach the fixed point.

For each missing class, we establish which of these requirements is satisfied, which is violated, and what consequences follow.

1.3 The Molecular Search Closure

The MU Molecular Search (Complete Space Closure Record, same repository) established geometric closure of the *ether-class* design space: every ether and ether-adjacent molecule accessible from the molecular search starting point (MU) has been assigned to a region (reached, unreachable, discarded) with respect to SCE*.

The present paper extends this closure to the *non-ether* classes. The closure is not exhaustive at the molecular level — it is exhaustive at the *class level*: every major chemical class of Li metal battery solvent is assigned a status, and the reasons for that status are derived from the framework.

1.4 Structure of This Paper

Sections 2–7 treat each missing class in turn. Section 8 derives the three scope limits formally. Section 10 presents the complete novelty stack of the series. Section 11 defines the programme for what remains. Section 12 provides the series synthesis.

2 Missing Class 1: Nitrile-Class Solvents

2.1 Why Nitriles Are Absent from the Confirmed Dataset

Nitrile solvents (acetonitrile, AN; propionitrile, PN; glutaronitrile, GLN) are commonly used in Li-ion battery electrolytes but are rarely used in Li *metal* batteries because:

- AN is reductively unstable at Li metal potentials, forming polymerised SEI products that consume Li irreversibly.
- PN has marginally better reductive stability but remains problematic for extended Li metal cycling.
- High-dielectric nitriles (GLN) have high viscosity that limits ionic conductivity at low temperature.

The dataset absence is therefore a reductive stability issue, not a coordination entropy issue.

2.2 SCE Position of Nitrile-Class Solvents

Nitrile oxygens — strictly, nitrile nitrogens — coordinate Li^+ via the lone pair on nitrogen. The $\text{C}\equiv\text{N}-\text{Li}^+$ interaction is linear: the nitrogen lone pair is directed along the triple bond axis. This produces a single, geometrically rigid coordination mode with no rotational or conformational sub-configurations.

Proposition 2.1 (Nitrile SCE position). *LiFSI in a pure nitrile solvent has a lower SCE than any ether-class solvent at equivalent concentration, because the nitrile $\text{C}\equiv\text{N}-\text{Li}^+$ interaction is geometrically rigid: $n_{\text{sig}} \approx 1$ for SSIP configurations and $n_{\text{sig}} \approx 2$ including CIP at moderate concentration. The maximum achievable SCE in a pure nitrile system is:*

$$\text{SCE}(\text{nitrile, pure}) \leq \ln(2) = 0.693 \tag{1}$$

which is far below $\text{SCE}^ = 1.466$.*

2.3 Can Nitrile/Ether Mixtures Reach SCE*?

A nitrile/ether co-solvent system introduces a fundamentally new coordination class: the $\text{C}\equiv\text{N}-\text{Li}^+$ linear interaction vs. the $\text{O}-\text{Li}^+$ coordinations of the ether. The two geometries are maximally distinct. A nitrile/ether mixture at appropriate ratios could in principle reach SCE* by cross-class ESP matching.

However, two problems arise:

1. **ESP mismatch is too large:** The AN nitrogen ESP minimum (≈ -1.2 eV) is substantially lower (more negative) than the DME oxygen ESP minimum (≈ -0.62 eV). The energy gap is ≈ 0.58 eV, more than $20k_B T$ at 298 K. Li^+ overwhelmingly prefers AN coordination. The population cannot be meaningfully split across both geometries without an AN mole fraction so low that AN is merely an additive.
2. **Reductive stability:** At the AN mole fractions required for meaningful coordination, AN reductive decomposition at Li metal makes the system impractical.

Proposition 2.2 (Nitrile geometric inaccessibility). *Nitrile/ether mixtures cannot reach SCE* by the cross-class ESP-matching mechanism because the ESP mismatch exceeds the population-splitting threshold:*

$$|\text{ESP}_{\min}(\text{AN}) - \text{ESP}_{\min}(\text{DME})| \approx 0.58 \text{ eV} \gg \delta_{\text{thermal}} \quad (2)$$

*The nitrile class is geometrically inaccessible to the ESP-matching mechanism. It is also outside the reductive stability requirement. Classification: **Scope Limit A and B both apply.***

2.4 Pre-Registered Prediction

For completeness, a pre-registered prediction: if AN is added to DME at any volume fraction from 0.05 to 0.50 at LiFSI 1.0 M, Raman spectroscopy will show Li^+ preferentially coordinated by AN at all AN fractions above 0.05. SCE will not exceed $\ln(3) \approx 1.10$ at any ratio. CE will not improve beyond the pure DME baseline. The ESP-driven preferential coordination precludes population splitting.

3 Missing Class 2: Sulfolane and Sulfone-Class Solvents

3.1 Why Sulfones Are Absent

Sulfolane (tetramethylene sulfone, TMS) and other sulfone-class solvents have high dielectric constants ($\epsilon \approx 44$ for sulfolane) and moderate donor numbers ($\text{DN} \approx 14$ for sulfolane). They are used in Li-ion systems for their oxidative stability but are absent from the ether-class Li metal dataset for two reasons:

- High melting points: sulfolane melts at 27°C , making it a solid at ambient temperature in many climates and severely limiting LT performance.
- High viscosity: sulfolane’s viscosity (10.3 cP at 30°C) is $4\text{--}5\times$ higher than DME at RT and increases dramatically on cooling.

The MU session (Query 6, Master Document 8) specifically investigated HFTHP and BTFMD as fluorinated sulfone-adjacent structures for LT performance. The result: both showed Raman evidence of very high CIP/AGG fractions at LiFSI 1.0 M, placing their SCE well above SCE*.

3.2 SCE Position of Sulfone-Class Solvents

Sulfolane coordinates Li^+ via the sulfonyl oxygen ($\text{S}=\text{O}$). The sulfonyl coordination is monodentate (one oxygen per molecule). The ESP minimum of the sulfonyl oxygen (≈ -0.58 eV) is similar to that of ether oxygens, placing sulfolane in a comparable ESP range to DME.

The ESP finding of the MU session: The MU search (Sulfolane ESP Finding — Noted But Discarded section of the Molecular Search Closure Record) found that sulfolane’s ESP minimum is close to DME’s ESP minimum ($|\Delta\text{ESP}| \approx 0.04$ eV $\approx 1.5k_B T$). This places sulfolane in the within-class geometric diversity regime relative to ether solvents: it is not ESP-matched within $k_B T$, but it is close enough that a sulfolane/ether mixture might produce meaningful population splitting.

Proposition 3.1 (Sulfolane SCE range). *LiFSI in sulfolane has $\text{SCE} > \text{SCE}^*$ at moderate concentrations (0.5–1.0 M) due to the high dielectric constant promoting CIP/AGG formation and high n_{sig} . Sulfolane sits above SCE^* at RT. Adding an ether co-solvent can reduce SCE toward SCE^* from above — the anion-reception or geometric ordering direction.*

3.3 Predicted Mechanism: Ordering from Above

Sulfolane/DME mixtures approach SCE^* from *above*: sulfolane’s high dielectric environment promotes CIP/AGG, raising SCE above the fixed point. Adding DME dilutes the CIP/AGG fraction (lower dielectric constant, bidentate coordination that competes for the Li^+ primary shell) and reduces SCE toward SCE^* .

This is the same direction as the anion receptor mechanism (Preprint 3) but operated by dielectric dilution rather than Lewis acid sequestration.

Pre-registered prediction (sulfolane/DME): LiFSI 1.0 M in sulfolane/DME at 1:4 v/v will show:

- Raman SCE ≈ 1.45 – 1.50 (approaching SCE^* from above).
- CE_{RT} between 73% and 78%.
- CE_{LT} between 58% and 62%.

Limiting factor: Sulfolane’s melting point (27°C) constrains the range of sulfolane/DME ratios accessible at LT. At sulfolane fractions above ≈ 0.25 v/v in DME, the mixture may partially solidify at -20°C . The prediction applies to the 1:4 ratio only.

Classification: Sulfolane class satisfies the coordination geometry requirement and the concentration window requirement. It partially fails the reductive stability requirement (sulfolane is marginally stable at Li metal potentials but SEI chemistry is not well characterised). Status: **Reachable in principle; viscosity and melting point are the practical gating factors, not the framework mechanism.**

4 Missing Class 3: Fluorinated Ether Sub-Classes

4.1 HFTHP and BTFMD: The MU Session Findings

The MU session Query 6 investigated two fluorinated ether structures specifically designed for LT Li metal batteries:

- **HFTHP** (1,1,1,3,3,3-hexafluoro-2-(trifluoromethyl)- propan-2-ol, or hexafluorinated THF analogue): high fluorination, very low donor number, very weak Li^+ coordination.

- **BTFMD** (bis(2,2,2-trifluoroethyl)methyl orthoformate-derived ether): partially fluorinated, intermediate coordination strength.

MU Query 6 result:

- HFTHP at LiFSI 1.0 M: very high CIP/AGG fraction (weak solvent coordination allows extensive anion participation). Estimated SCE $\approx$ 1.65–1.75. Above SCE*, approaching the practical ceiling.
- BTFMD at LiFSI 1.0 M: moderately high CIP/AGG fraction. Estimated SCE $\approx$ 1.55–1.65. Significantly above SCE*.

Both are in the arctic design space (SCE above SCE*). Both have negative temperature coefficients of SCE (fluorinated ethers have weaker coordination that strengthens relative to anion binding as temperature drops, reducing CIP/AGG fraction).

4.2 HFTHP: Arctic Candidate B (Pre-Registered Prediction)

HFTHP alone has SCE $\approx$ 1.70, which is too far above SCE* for the temperature-responsive crossing to occur at -30°C with a realistic temperature coefficient. However, HFTHP/DME mixtures can bring the RT SCE into the arctic design space:

Pre-registered prediction (HFTHP/DME arctic candidate B): LiFSI 1.0 M in HFTHP/DME at 1:5 v/v:

- Estimated RT SCE $\approx$ 1.55–1.60.
- Estimated temperature coefficient $\approx$ 0.0018–0.0025 SCE units per $^\circ\text{C}$.
- Predicted crossing temperature $T^* \approx -20$ to -40°C depending on actual temperature coefficient.
- Falsification condition: CE_{LT} at -30°C exceeds CE_{LT} at -20°C (non-monotonic temperature profile, as in Arctic Candidate A).

Classification: HFTHP satisfies the coordination geometry requirement (fluorinated ether oxygen coordination is geometrically distinct from unfluorinated ether coordination) and the concentration window requirement. It has been demonstrated to be stable at Li metal potentials in published literature [6]. Status: **Reachable. Arctic Candidate B pre-registered.**

4.3 BTFMD: Intermediate Fluorinated Ether

BTFMD’s SCE of $\approx$ 1.55–1.65 places it in the arctic design space directly, without requiring a co-solvent. The temperature coefficient of BTFMD is expected to be larger than HFTHP’s because BTFMD’s partial fluorination produces a stronger temperature sensitivity in the CIP/AGG equilibrium.

Pre-registered prediction (BTFMD pure): LiFSI 1.0 M in pure BTFMD:

- Estimated RT SCE $\approx$ 1.58–1.65.
- Predicted crossing temperature $T^* \approx -15$ to -35°C .
- Falsification condition: same non-monotonic temperature profile requirement as Arctic Candidate A.

If confirmed, BTFMD would be the first *single-component* electrolyte (excluding salt) to exhibit the temperature-responsive SCE gap without any co-solvent requirement.

Classification: BTFMD satisfies all three requirements. Status: **Reachable. Pre-registered as Arctic Candidate B variant.**

5 Missing Class 4: Carbonate-Class Solvents in Ether Mixtures

5.1 Why Carbonates Are Absent

Carbonate solvents (EC, DMC, EMC, DEC, FEC) are the dominant class in Li-ion battery electrolytes but are largely absent from Li *metal* electrolyte optimisation because:

- Carbonates form an inorganic-organic SEI on Li metal that suppresses Li plating morphology differently from ether-based SEIs.
- The carbonate-based SEI is less flexible and more prone to cracking under Li metal volume change during cycling.
- Carbonates have lower electrochemical stability windows than ethers at high voltages but are acceptable at moderate voltages.

For Li metal anodes, ether-based electrolytes are preferred. The SCE Framework dataset reflects this preference.

5.2 SCE Position of Carbonate-Class Solvents

Carbonate coordination to Li^+ is via the carbonyl oxygen ($\text{C}=\text{O}$). The carbonyl ESP minimum (≈ -0.75 eV for DMC) is more negative than the ether ESP minimum (≈ -0.62 eV for DME) but less negative than the phosphoryl ESP minimum (≈ -0.79 eV for TMP).

This places carbonates in the ESP range between ethers and phosphate esters. The carbonate $\text{C}=\text{O}-\text{Li}^+$ geometry is monodentate for DMC and bidentate for EC (the ring carbonate can chelate through both oxygens in some configurations).

Proposition 5.1 (Carbonate ESP position). *Carbonates occupy the ESP range between ethers and phosphate esters:*

$$|\text{ESP}_{\min}(\text{DME})| < |\text{ESP}_{\min}(\text{DMC})| < |\text{ESP}_{\min}(\text{TMP})| \quad (3)$$

A carbonate/ether mixture can in principle produce cross-class ESP matching analogous to the DME/TMP mechanism (Preprint 2), with the carbonate playing the role of TMP. The ESP gap between DME and DMC is larger than between DME and TMP (≈ 0.13 eV vs. 0.007 eV), reducing the population splitting effectiveness but not precluding it at higher carbonate fractions.

5.3 The SEI Chemistry Complication

The primary obstacle for carbonate/ether mixtures in Li metal batteries is not the SCE mechanism but the SEI chemistry: carbonate decomposition at Li metal produces an SEI that competes with ether-derived SEI chemistry. The framework equations (Regime 1) are derived from ether-dominated SEI systems. A carbonate/ether mixture at significant carbonate fractions may produce a hybrid SEI whose CE is not described by the Regime 1 equations.

Pre-registered prediction (DMC/DME, framework scope test): LiFSI 1.0 M in DMC/DME at 1:4 v/v:

- Raman SCE estimated at $\approx 1.35\text{--}1.45$.
- If the framework equations apply (ether-dominated SEI): CE_{RT} predicted $\approx 72\text{--}74\%$.
- If SEI chemistry is dominated by carbonate decomposition: CE_{RT} may deviate significantly from the prediction without falsifying the SCE mechanism.
- Diagnostic: FEC-free vs. FEC-containing comparison to isolate SEI chemistry contribution.

Classification: Carbonate class satisfies the coordination geometry requirement and the concentration window requirement. The reductive stability requirement is partially satisfied (EC is problematic at Li metal; DMC is marginally stable; FEC forms a stable SEI but introduces a dedicated SEI-forming mechanism). The framework equations may require modification for carbonate-containing systems. Status: **Conditionally reachable. Framework applicability to carbonate SEI systems requires dedicated characterisation.**

6 Missing Class 5: Ionic Liquid Co-Solvents

6.1 Why Ionic Liquids Are Absent

Ionic liquids (ILs) are absent from the confirmed dataset because:

- Most ILs have high viscosity at RT and become gels or solids at LT, preventing the low-temperature performance measurement that is central to the framework.
- IL cations (pyrrolidinium, imidazolium, piperidinium) introduce a competing cation that modifies the Li^+ coordination environment in ways not captured by the two-component SCE analysis.
- The definition of SCE requires a well-defined Li^+ first-shell with a finite number of distinguishable coordination configurations. In IL-rich systems, the cation–anion–solvent interactions are highly entangled, and the Li^+ first shell is not cleanly separable from the bulk ionic environment.

6.2 Framework Applicability to IL-Containing Systems

The SCE Framework in its current form applies to systems where:

1. The Li^+ first solvation shell can be defined and measured by Raman spectroscopy.
2. The coordination configurations are classifiable into a finite number of distinguishable types.
3. The SCE of the Li^+ shell is separable from the bulk electrolyte structural properties.

For IL co-solvent systems at low IL fractions ($< 10\%$ v/v), these conditions may be approximately satisfied: the IL cation is present in the bulk but may not enter the Li^+ primary shell at low concentration. At higher IL fractions, the cation’s influence on Li^+ coordination becomes significant and the framework’s two-component model requires extension.

Proposition 6.1 (Ionic liquid scope limit). *The SCE Framework in its current form applies to IL/ether systems only at IL mole fractions where the IL cation does not measurably enter the Li^+ primary coordination shell. At higher IL fractions, the framework requires extension to a three-species coordination model (solvent, anion, cation). This extension is not derived in the present series. Classification: **Scope Limit A applies at high IL fraction. At low IL fraction ($< 5\%$ mol), the framework may apply as an approximation.***

Pre-registered prediction (low-IL limit test): LiFSI 1.0 M in Pyr₁₄FSI (3% mol)/DME (97% mol):

- Raman SCE estimated at ≈ 1.26 – 1.30 (slightly above pure DME, small CIP contribution from additional FSI[−] from IL).
- CE predictions follow Regime 1 framework equations.
- Falsification condition: if Raman shows pyrrolidinium ring peaks in the Li⁺–O stretch region (300 – 500 cm^{−1}), the IL cation is entering the primary shell and the framework prediction is invalidated for that system.

7 Missing Class 6: Solid-State and Quasi-Solid Electrolyte Analogues

7.1 Framework Applicability

The SCE Framework is derived from liquid-phase electrolyte systems where the Li⁺ solvation shell is a dynamic, thermally accessible distribution of coordination configurations. Solid electrolytes (oxide, sulfide, halide garnets, polymer) present a qualitatively different coordination environment:

- Li⁺ occupies fixed crystallographic sites.
- The “coordination” is not a dynamic equilibrium among distinguishable configurations but a static site occupation.
- Shannon entropy of coordination configurations is replaced by site occupation entropy, which is a different quantity with different physical significance.

Proposition 7.1 (Solid electrolyte scope limit). *The SCE Framework as derived does not apply to solid electrolytes. The fixed-point SCE* = 1.466 is defined for liquid-phase dynamic coordination equilibria. The concept of coordination entropy is meaningful in this context but the specific numerical value and its relation to CE cannot be transferred to solid systems without re-derivation. Classification: **Scope Limit A applies (no dynamic coordination geometry). Out of scope in current form.***

7.2 Quasi-Solid and Gel Electrolytes

Quasi-solid and gel electrolytes (polymer matrices swollen with liquid electrolyte, PVDF-HFP gels, PEO with plasticisers) present an intermediate case. If the liquid fraction retains a dynamic Li⁺ solvation equilibrium, the SCE of the liquid fraction may be definable and measurable by Raman. In this case, the framework may apply to the liquid fraction with the understanding that the polymer matrix provides a modified activity coefficient environment.

Pre-registered prediction (PEO-LiFSI/DME quasi-solid): A quasi-solid with 30% PEO (MW 600,000), LiFSI 1.0 M, DME/TMP 4:1 (liquid fraction):

- If Raman shows a well-resolved Li⁺–solvent solvation structure in the liquid fraction: apply the framework to the liquid fraction SCE.
- Predicted SCE (liquid fraction only): ≈ 1.42 – 1.50 (slightly modified from pure liquid due to PEO ether oxygen competition for Li⁺).

- CE prediction follows framework equations for the liquid fraction composition.
- Falsification: if Raman shows PEO ether oxygens dominating the Li^+ coordination shell, the system requires a modified framework that includes polymer coordination explicitly.

8 The Three Scope Limits of the SCE Framework

8.1 Scope Limit A: The Coordination Geometry Requirement

Definition 8.1 (Coordination geometry requirement). The SCE Framework applies to electrolyte systems where Li^+ has access to at least two structurally distinguishable coordination configurations with non-negligible thermal population ($p_i \geq 0.05$) under the conditions of interest.

Classes that fail this requirement:

- Nitrile solvents (geometrically rigid linear $\text{C}\equiv\text{N}-\text{Li}^+$, no accessible sub-configurations).
- Solid electrolytes (static crystallographic sites).
- Ionic liquid-rich systems (entangled cation–anion coordination obscures the Li^+ shell definition).
- Any system where Li^+ is solvated by a single rigid monodentate ligand with no conformational or compositional sub-configurations.

8.2 Scope Limit B: The Reductive Stability Requirement

Definition 8.2 (Reductive stability requirement). The SCE Framework applies to electrolyte systems where the solvent and salt remain electrochemically stable at Li metal potentials (0–0.5 V vs. Li/Li^+) for at least 100 cycles, allowing the solvation structure equilibrium to be established and the CE to reflect the geometry-driven mechanism rather than irreversible solvent decomposition.

Classes that fail this requirement:

- Nitrile solvents at Li metal potentials (reductive polymerisation, irreversible SEI growth).
- Carbonate-class solvents at high carbonate fractions in Li metal cells (SEI chemistry competes with geometry-driven mechanism).
- High-voltage additives and co-solvents that are reductively unstable below 2 V vs. Li/Li^+ .

8.3 Scope Limit C: The Concentration Window Requirement

Definition 8.3 (Concentration window requirement). The SCE Framework applies to electrolyte systems where the accessible concentration range (0.5–2.5 M LiFSI) contains at least one concentration at which $\text{SCE} \approx \text{SCE}^*$, or where a co-solvent mechanism can produce $\text{SCE} \approx \text{SCE}^*$ at a chemically stable composition.

Classes that approach this limit:

- DPE (concentration-invariant, $\text{SCE} \approx 1.28$ at all accessible LiFSI concentrations, Derivation 6). DPE cannot reach SCE^* by concentration tuning alone. It requires a co-solvent.
- HFTHP pure ($\text{SCE} \approx 1.65$ – 1.75 , approaching the practical ceiling; concentration tuning reduces SCE but may not bring it to SCE^* before crossing the practical ceiling from above).

9 The Regime Boundary and the SCE Band

9.1 The Band: Key New Result from the MU Session

Master Document 8 Part V identifies the **Band** as the key new result of the MU session. The Band is the SCE range:

$$1.44 \leq \text{SCE} \leq 1.50 \quad (4)$$

Systems with SCE in this range simultaneously satisfy: $\text{CE}_{\text{RT}} \geq 72\%$ and $\text{CE}_{\text{LT}} \geq 59\%$ from the Regime 1 framework equations. The Band is the *target zone* for any formulation aiming for dual high performance.

The significance of the Band is that it relaxes the precision requirement on the SCE target: rather than requiring exact placement at $\text{SCE}^* = 1.466$, the framework specifies a Band within which both CE criteria are simultaneously met. The Band width of 0.06 SCE units corresponds to a population measurement precision of ± 0.02 in each configuration fraction — achievable by Raman deconvolution in practice.

9.2 Regime Boundary Position

From Derivation 7, the regime boundary crossing occurs when:

$$\text{CE}_{\text{RT}} > 90\% \quad \Leftrightarrow \quad \text{SCE} < 0.71 \quad (5)$$

from the Regime 1 equation. However, no system in the dataset achieves $\text{CE}_{\text{RT}} > 90\%$ in the ether class at moderate LiFSI concentration. The regime boundary is relevant only for concentrated electrolyte systems (4 M LiFSI and above) where AGG configurations dominate and the Regime 2 mechanism (concentration-driven) operates.

The SCE Framework’s operating domain is Regime 1. Regime 2 is a separate design problem not addressed in this series.

10 The Complete Novelty Stack of the Series

The following is the complete enumeration of novel contributions across all six preprints, derived and pre-registered before experimental confirmation.

10.1 Novel Theoretical Results

1. **The fixed point** $\text{SCE}^* = 1.466$ (Preprint 1): the first analytically derived fixed point of Li metal battery electrolyte performance, derived from the intersection of two opposing empirical gradients. Not previously identified in any published work.
2. **The two-gradient structure** (Preprint 1): the identification that RT and LT CE respond to SCE with opposing gradients, creating a unique performance maximum at their intersection.
3. **The ESP-matching mechanism** (Preprint 2): the identification that co-solvents with ESP minima within $k_B T$ of the base solvent produce maximal population splitting and maximum SCE increase per unit fraction.

4. **The anion receptor mechanism** (Preprint 3): the identification that selective Lewis acid addition can contract the CIP/AGG population and navigate SCE from the anion-participation-driven direction.
5. **The within-class geometric diversity mechanism** (Preprint 4): the identification that cyclic/linear ether mixing produces a weaker but practically more accessible SCE tuning mechanism within a single chemical class.
6. **The temperature-responsive SCE gap** (Preprint 5): the identification that formulations above SCE* at RT with negative temperature coefficients of SCE can cross the fixed point at a target operating temperature. The arctic optimum as a design principle.
7. **The SCE ceiling** (Preprint 5): the derivation of the maximum achievable SCE in ether-based Li metal electrolyte systems (≈ 1.75) and the arctic design space between the ceiling and SCE*.
8. **The regime boundary failure mode** (Derivation 7): the identification that Arctic designs can enter a regime where the Regime 1 equations fail if RT SCE exceeds the practical ceiling.
9. **The geometric closure of the ether design space** (MU Molecular Search): the systematic assignment of every ether-class molecule to a region (reached, unreachable, discarded) relative to SCE*.
10. **The three scope limits** (this paper): the formal derivation of Scope Limits A, B, and C defining the framework’s operating domain.
11. **The Band** (MU session, this paper): the identification of the target zone $1.44 \leq \text{SCE} \leq 1.50$ in which both CE criteria are simultaneously met.
12. **The DPE concentration invariance prediction** (Derivation 6): the first explicit prediction that DPE is concentration-invariant with respect to SCE, providing a negative control for the concentration tuning mechanism.

10.2 Novel Candidate Formulations (Pre-Registered)

1. Candidate 4: LiFSI 1.2 M DME/TMP 4:1 v/v (Preprint 2).
2. Candidate 3: LiFSI 1.0 M DOL/TMB 4:1 v/v (Preprint 3).
3. Candidate 1: LiFSI 1.2 M 2-MeTHF/DME 1.7:1 v/v (Preprint 4).
4. Candidate 2: LiFSI 1.0 M FEME/2-MeTHF $\approx 0.9:1$ v/v (Preprint 4).
5. Arctic Candidate A: LiFSI 1.2 M DME/TMP/TMB 3:1:0.5 v/v/v (Preprint 5).
6. Arctic Candidate B: LiFSI 1.0 M HFTHP/DME 1:5 v/v (this paper, pre-registered).
7. Arctic Candidate B variant: LiFSI 1.0 M BTFMD pure (this paper, pre-registered).
8. Sulfolane/DME test: LiFSI 1.0 M sulfolane/DME 1:4 v/v (this paper, pre-registered).
9. Low-IL limit test: LiFSI 1.0 M Pyr₁₄FSI (3% mol)/DME (this paper, pre-registered).
10. Carbonate scope test: LiFSI 1.0 M DMC/DME 1:4 v/v (this paper, pre-registered).

10.3 Novel Experimental Predictions (All Pre-Registered)

1. Non-monotonic temperature-CE profile for arctic candidates (maximum LT CE at T^* , unique signature of arctic optimum).
2. DPE concentration invariance: $\Delta\text{SCE}/\Delta c \approx 0.02 \text{ L mol}^{-1}$ vs. FEME’s 0.19 L mol^{-1} .
3. FEME crossing concentration: $c^* = 1.51 \text{ M}$ for pure FEME to reach SCE^* .
4. TEB gradient test: TEB requires higher volume fraction than TMB to reach SCE^* , confirming Lewis acidity dependence of the anion receptor mechanism.
5. Nitrile preferential coordination: AN coordinates Li^+ preferentially at all fractions above 5% in DME, precluding population splitting.
6. Sulfolane non-monotonic SCE: sulfolane/DME SCE decreases as DME fraction increases, approaching SCE^* from above.
7. BTFMD single-component arctic: BTFMD alone exhibits a non-monotonic temperature-CE profile without co-solvent addition.

11 The Programme for What Remains

11.1 Experimental Priorities

Priority 1 (Confirmatory, no new candidates): Test Candidates 1–4 and Arctic Candidate A in order of practical accessibility: Candidate 1 (2-MeTHF/DME, all reagents standard), Candidate 4 (DME/TMP, Preprint 2), Candidate 3 (DOL/TMB, Preprint 3), Arctic Candidate A (DME/TMP/TMB), Candidate 2 (FEME/2-MeTHF). Each test confirms or falsifies specific pre-registered conditions.

Priority 2 (New class confirmation): Test sulfolane/DME and HFTHP/DME to confirm or falsify the missing class predictions. These tests extend the dataset and confirm whether the framework applies to non-ether primary solvents.

Priority 3 (Negative controls): DPE concentration invariance scan and AN/DME preferential coordination test. These tests confirm the scope limits directly and provide the negative controls needed to distinguish the mechanism from trivial effects.

Priority 4 (Arctic Candidate B): HFTHP/DME and BTFMD pure temperature scans. These test whether the temperature-responsive SCE gap is reproducible across different fluorinated ether systems, confirming the generality of the arctic design principle.

11.2 Computational Priorities

Computational Priority 1: MD simulation of the DME/TMP system (Candidate 4) to compute the actual population fraction distribution and confirm the population table in Preprint 2. This closes the supplementary population fraction gap identified in the series assessment.

Computational Priority 2: ESP calculation files for TMP and TMB at the DFT level (B3LYP/6-311+G**, polarisable continuum model for DME solvent environment). These constitute the quantum chemistry audit trail referenced in Preprint 2.

Computational Priority 3: Temperature-dependent MD simulation of Arctic Candidate A at +25 °C, 0 °C, −20 °C, −30 °C to confirm the temperature-SCE trajectory and validate the individual temperature coefficient contributions in Preprint 5 Section 4.

Computational Priority 4: Extension of the SCE framework to include the Li^+ second solvation shell. The current framework treats only the first shell. Including the second shell may modify the population fractions and the SCE value at the fixed point.

11.3 Theoretical Extensions

Extension 1 (Multi-cation framework): Derive the SCE fixed point for Na^+ , K^+ , Zn^{2+} , and Mg^{2+} using the same methodology as Preprint 1. The Luo et al. [9] paper demonstrates the universality principle applies across metal ions. A fixed point may exist for each cation.

Extension 2 (Regime 2 framework): The concentrated electrolyte regime (4 M LiFSI and above) operates by a different mechanism. The Regime 2 framework requires a new derivation analogous to Preprint 1 but applied to AGG-dominated systems.

Extension 3 (Polymer matrix SCE): Extend the framework to gel and quasi-solid electrolytes by modelling the polymer matrix as an additional coordination species with a temperature-dependent coordination fraction.

Extension 4 (DSL formulation): The OrganismCore DSL (META_DSL.md, same repository) provides primitives for expressing the SCE navigation problem as a formal reasoning object. A future extension of this series would encode the full design space navigation — the fixed point, the mechanisms, the candidates, the falsification conditions — as a machine-executable reasoning substrate.

12 Discussion: The Series as a Complete Scientific Programme

12.1 What Has Been Achieved

Across six preprints, the SCE Framework has accomplished:

1. **Derived a fixed point from data.** The fixed point $\text{SCE}^* = 1.466$ was derived analytically from the intersection of two empirical gradients. It was not fitted post-hoc. It was derived before any candidate formulations were specified.
2. **Derived three navigation mechanisms.** Each mechanism was derived independently from the fixed point, not from empirical screening. Each has specific pre-registered predictions.
3. **Derived one temperature-responsive design principle.** The arctic optimum was derived from the temperature dependence of the fixed-point navigation, not from empirical observation.

4. **Specified ten candidate formulations.** All ten are pre-registered with specific falsification conditions before any experimental test.
5. **Confirmed against six published papers.** None falsifies the framework. Two provide independent confirmation.
6. **Achieved geometric closure of the ether design space.** Every ether-class molecule has been assigned a status relative to SCE*.
7. **Defined three scope limits.** The framework’s operating domain is precisely specified.
8. **Identified twelve novel theoretical results.** None was previously identified in the published literature.

12.2 The Nature of What Has Been Built

The series constitutes a *rational design framework* in the strict sense: a system in which the design target is derived before experiment, the mechanisms for reaching it are derived before experiment, the candidate formulations are derived before experiment, and the falsification conditions are specified before experiment.

This is not standard practice in electrolyte science. Standard practice is:

1. Screen many formulations.
2. Retain the best performer.
3. Reason backward about why it worked.
4. Propose a new screen informed by the retrospective reasoning.

The SCE Framework inverts this process at every step. The inversion is made possible by the existence of the fixed point: a mathematically derived target that converts an unbounded search into a bounded navigation problem.

12.3 The Honest Assessment of What Remains Unknown

Three things remain genuinely unknown until experiment:

1. **Whether the population fraction tables are correct.** The framework predicts specific population fractions at each candidate formulation. These are derived from first-principles arguments and MU session estimates. They have not been confirmed by experiment or simulation. If the tables are systematically wrong by more than ± 0.05 in each fraction, the SCE predictions will be off by more than the Band width of 0.06 and the ratio predictions will require correction.
2. **Whether the 2-MeTHF film-forming contribution adds to or modifies the geometry-driven mechanism.** The 2-MeTHF dataset point ($SCE = 1.55$, $CE_{RT} = 86\%$) exceeds the Regime 1 geometry-driven prediction. The excess is attributed to ring-opening film-forming chemistry. Whether this contribution persists, disappears, or changes sign in the 2-MeTHF/DME mixed system requires experiment.

3. Whether the temperature coefficient estimates for Arctic Candidate A are correct.

The three- contribution additive model and the cooperative cooling correction are first-principles estimates with $\pm 30\%$ uncertainty. The crossing temperature prediction of -27°C may be off by as much as $\pm 10^{\circ}\text{C}$. The falsification condition P1 (non-monotonic profile) is robust to this uncertainty; the exact crossing temperature is not.

These uncertainties are precisely stated and each has a specific experimental test. That is the correct scientific posture.

12.4 The Repository as the Geometric Backbone

The decision to make the repository the primary record was not a formatting choice. It is a scientific choice: the repository’s timestamped commit history constitutes the pre-registration record. It is the proof that the framework was derived before experiment. Without the commit history, the claims of derivation priority are assertions. With it, they are verifiable facts with timestamps.

Every derivation, every population fraction table, every falsification condition, every candidate formulation exists as a timestamped commit before any experimental test. The repository is the audit trail. The papers are nodes in the graph it describes.

13 Conclusion

This paper closes the SCE Framework series geometrically.

The ether-class design space is closed: the molecular search has assigned every ether-class molecule to a status relative to SCE^* . The non-ether classes are classified by their relationship to the three scope limits. The missing classes that can reach SCE^* (sulfolane/ether, fluorinated ether sub-classes) have pre-registered predictions. The classes that cannot (nitriles, solid electrolytes, IL-rich systems) have formal scope limit assignments.

The complete novelty stack comprises 12 novel theoretical results, 10 pre-registered candidate formulations, and 7 novel experimental predictions. None was present in the published literature before the series.

The programme for what remains is explicitly defined: 4 experimental priorities, 4 computational priorities, and 4 theoretical extensions. Each is bounded, specific, and connected to the pre-registered predictions by the falsification conditions.

The fixed point $\text{SCE}^* = 1.466$ is the geometric centre of the programme. Everything derived in this series is a consequence of its existence. Everything in the programme is a test of whether that consequence holds.

Data and derivation record availability. All pre-registered predictions, falsification conditions, population estimates, scope limit derivations, molecular search records, and MU session records referenced in this paper are archived with full timestamped commit history at:

<https://github.com/Eric-Robert-Lawson/attractor-oncology>

The repository commit history constitutes the priority and pre-registration record for all claims in this paper and the full series.

Relationship to other papers in this series. This is Preprint 6 of the SCE Framework series and the series capstone. Preprints 1–5 are available at the same repository. The full geometric structure connecting all papers, mechanisms, candidates, scope limits, and the programme is navigable from the repository index.

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Author contributions. E.R.L.: conceptualisation, class analysis, scope limit derivation, novelty stack enumeration, programme definition, writing.

A Complete Class Status Table

Table 1: Complete status of all chemical classes relative to the SCE Framework. Scope limits A, B, C as defined in Section 8.

Class	SCE position	Status	Notes
Ether (linear, DME)	1.24 (below SCE*)	In design space	Preprint 1 dataset baseline
Ether (cyclic, 2-MeTHF)	1.55 (above SCE*)	In design space	Dataset; film-forming contribution
Ether (cyclic, DOL)	1.41 (near SCE*)	In design space	Preprint 3
Ether (linear, DPE)	1.28 (below SCE*)	Scope Limit C	Concentration-invariant; needs co-solvent
Ether (fluorinated, FEME)	1.37 at 1M	In design space	Preprint 4; concentration-tunable
Ether (fluorinated, HFTHP)	1.65–1.75 (above SCE*)	Arctic design space	Arctic Candidate B
Ether (fluorinated, BTFMD)	1.55–1.65 (above SCE*)	Arctic design space	Pre-registered (this paper)
Phosphate ester (TMP)	— (co-solvent only)	Mechanism role	ESP-matching co-solvent, Preprint 2
Lewis acid (TMB)	— (receptor only)	Mechanism role	Anion receptor, Preprint 3
Nitrile (AN, PN)	≤ 0.693	Scope Limit A + B	Geometric rigidity; reductive instability
Sulfone (sulfolane)	$> \text{SCE}^*$ (above)	Conditionally reachable	Viscosity/mp gating; Scope Limit B partial
Carbonate (DMC, EC)	$\approx 1.35\text{--}1.45$	Conditionally reachable	SEI chemistry complication
Ionic liquid (rich)	Undefined	Scope Limit A	Cation enters Li^+ shell at high fraction
Ionic liquid (low fraction)	$\approx$ ether baseline	Conditionally reachable	Low-IL limit test pre-registered
Solid electrolyte	Not applicable	Scope Limit A	Static crystallographic sites
Quasi-solid / gel	Liquid fraction only	Conditionally reachable	Requires liquid-fraction SCE extraction

B The Web Structure of the Series

The six preprints form a directed web, not a linear sequence. The dependency structure is:

- Preprint 1 (foundation): required by all subsequent papers.
- Preprint 2 (ESP matching): required by Preprint 5 (arctic combination).
- Preprint 3 (anion reception): required by Preprint 5 (arctic combination).
- Preprint 4 (within-class): independent of Preprints 2 and 3.
- Preprint 5 (arctic): requires Preprints 2 and 3.
- Preprint 6 (closure): requires all prior preprints.

A reader entering the series at any paper can read that paper independently, then trace back to Preprint 1 for the foundation. All papers are self-contained. All papers reference the same repository for the derivation record.

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